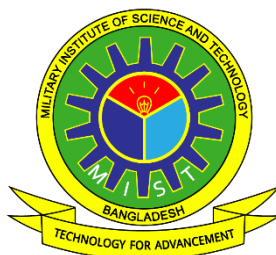


Military Institute of Science and Technology

Department of Electrical Electronic and Communication Engineering

EECE – 222: Electrical Service Design & CAD Laboratory

Course Code: EECE-222



Project Report

Group-A9

Topic- Auditorium Floor Plan & Electrical lines with proper calculation

Name	ID
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Date of Submission	December 2,2023
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Index

Sl. No	Topic	Page No
1	Introduction	1
2	Figures • Figure 1 – Floorplan • Figure 2 – Floor plan with all fittings and fixtures • Figure 3 – Conduit layout • Figure 4 – SLD	
3	Calculation	
4	Estimation of Total Cost	
5	Conclusion	
6	Reference	

Auditorium Floor Plan & Electrical lines with proper calculation

Abstract

This document presents a formal description of the architectural design assigned, specifically an Auditorium. Auditoriums serve as pivotal spaces for hosting diverse events, performances, lectures, and gatherings.

Introduction

The focus of this architectural assignment is an **Auditorium**. These spaces are integral components within infrastructures, facilitating a wide array of events and presentations. They are meticulously designed to ensure optimal comfort and functionality for both audience members and presenters alike.

Floor Plan Description

The floor plan of the auditorium embodies a structured layout that harmonizes with its intended functionalities:

1. **Main Hall:** The central area of the auditorium where audience seating is strategically arranged to optimize visibility and acoustics. The seating configuration is tailored to provide a clear line of sight to the stage or presentation area, ensuring an engaging experience for attendees.
2. **Stage Area:** Positioned as a focal point, the stage is equipped with sophisticated lighting, sound systems, and customizable backdrops, catering to diverse event requirements. The stage dimensions accommodate speakers, performers, or presenters, offering ample space and amenities for effective content delivery.
3. **Technical Booths & Media Center:** Dedicated spaces housing technical equipment such as lighting controls, sound systems, projectors, and AV gear. These booths are strategically positioned to facilitate seamless operations during events, ensuring optimal technical support.

4. Access Ways: Emphasis is placed on well-designed entrances/exits, aisles, and pathways to ensure smooth audience and staff navigation during events. Safety and accessibility standards are rigorously adhered to within the design layout.

Assignment Details

The floor plan of this Auditorium spans an area of 5000 square feet, offering versatility in accommodating moderate to large audiences. This space is amenable to various seating configurations, ensuring a comfortable and engaging experience for attendees across a spectrum of events and presentations.

Our auditorium is basically divided in **four** segments:

- Main Hall
- Stage Area
- Back Stage
- General Discussion and Entrance

Length = 114 ft. 4 inches

Width= 42 ft. 3 inches

The Main Hall, with its dimensions of **39 feet** in width and **43 feet** in length, stands as the primary space dedicated to accommodating auditorium seating within our single-story establishment. The provision of an optimal mounting height for superior lighting is pivotal to enhance visibility and create a conducive environment for various events and presentations.

Within the backstage area, our focus remains on providing ample room and clarity to ensure the comfort and convenience of artists and performers. It's imperative that this space is adequately sized not only for their comfort but also to house an efficient air conditioning system, catering to the varying needs of different performances and events.

Moreover, the design emphasizes a spacious and accessible general discussion area that serves as a hub for all attendees. This space is envisioned to foster meaningful interactions and engagement among participants, creating an inclusive environment that encourages dialogue and collaboration.

Additionally, the entrance is designed to be well-lit and spacious, creating a welcoming ambiance that sets the tone for the overall auditorium experience. Its ample space aims to accommodate the flow of visitors seamlessly, ensuring a smooth and pleasant entry into the premises.

Figures:

Figure 1 – Floorplan

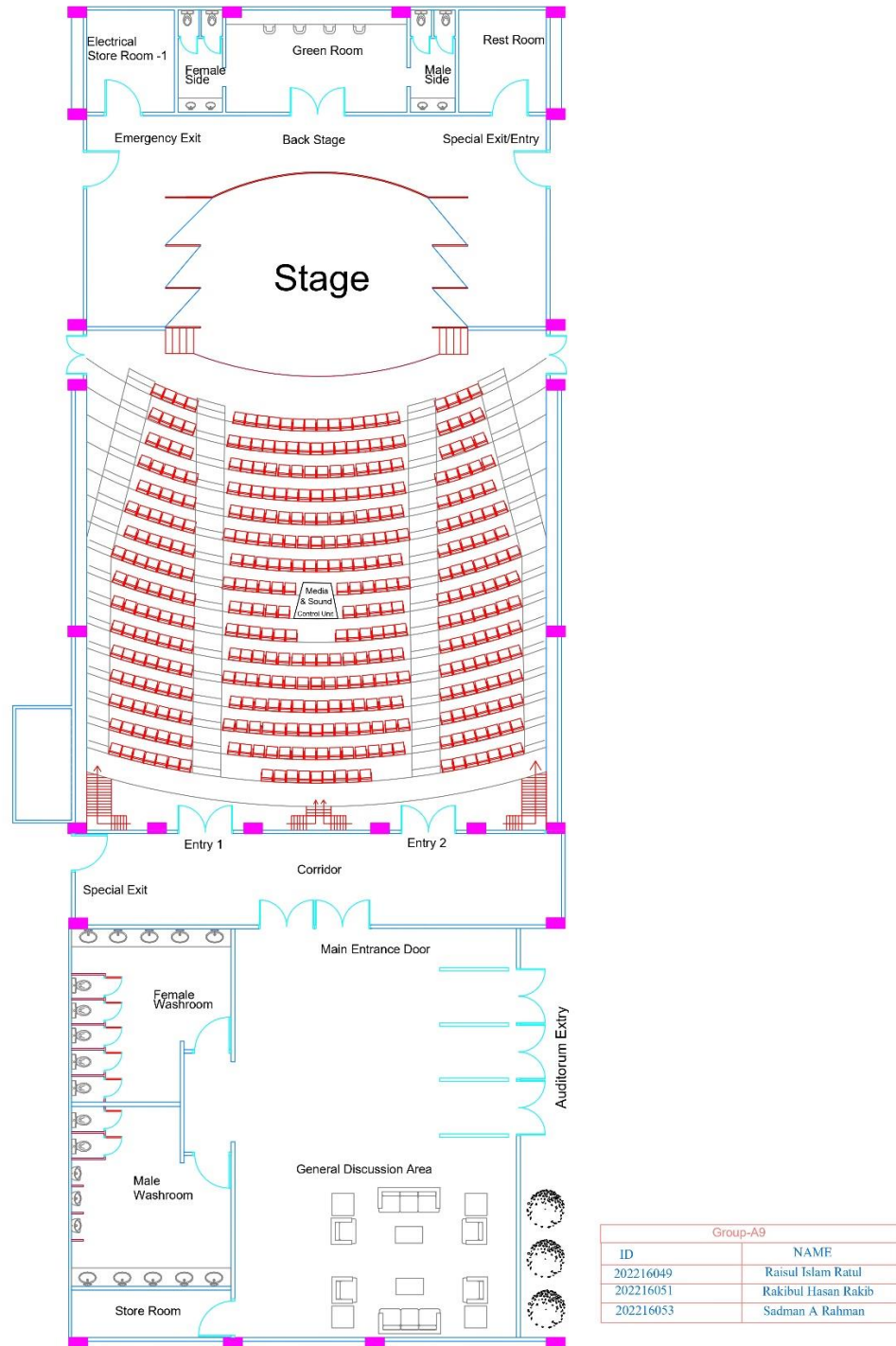


Figure 1– Floor plan

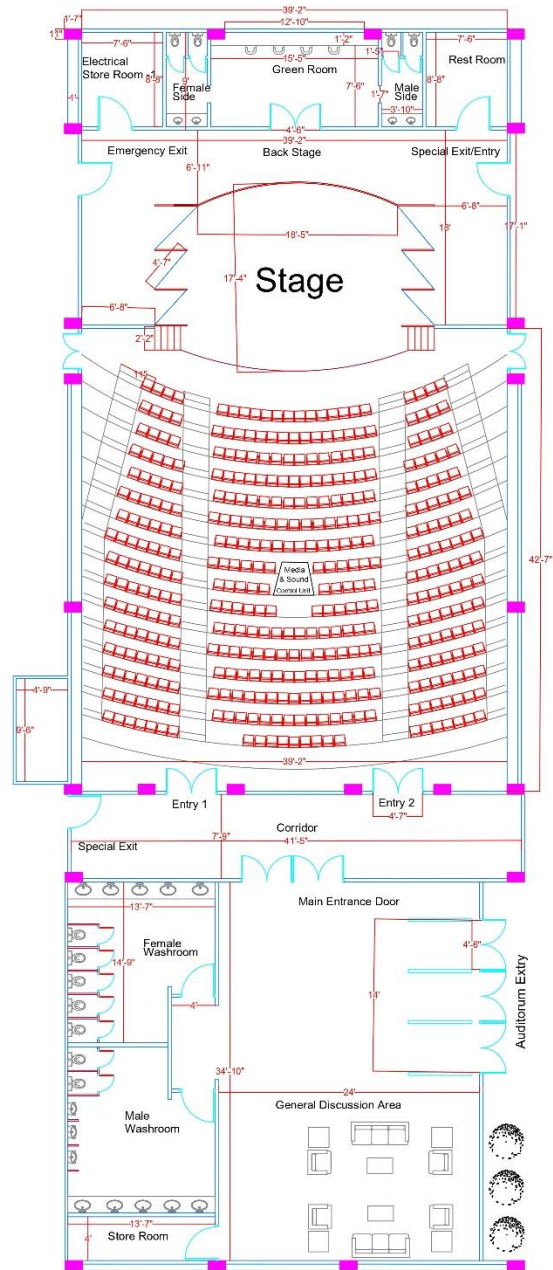


Figure 2 Floor Plan with Dimension

Figure 2 – Floor plan with all fittings and fixtures placed in appropriate position

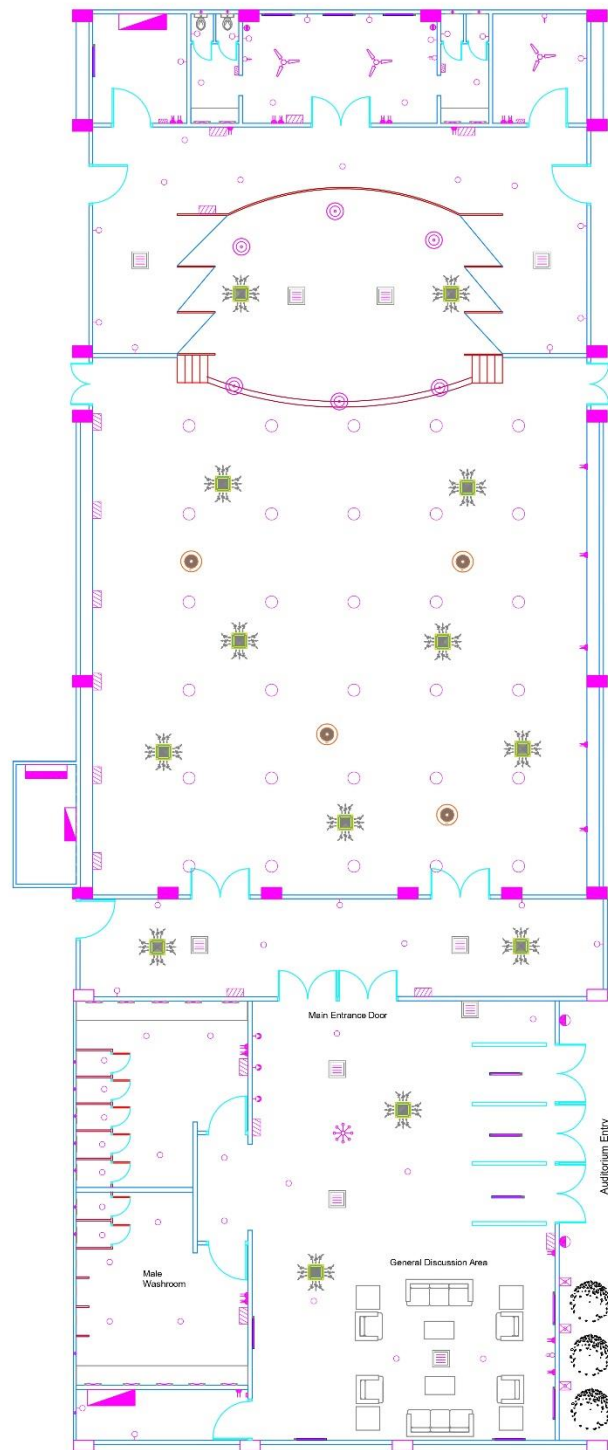


Figure 3 Fitting -Fixture

Figure 3 – Conduit layout

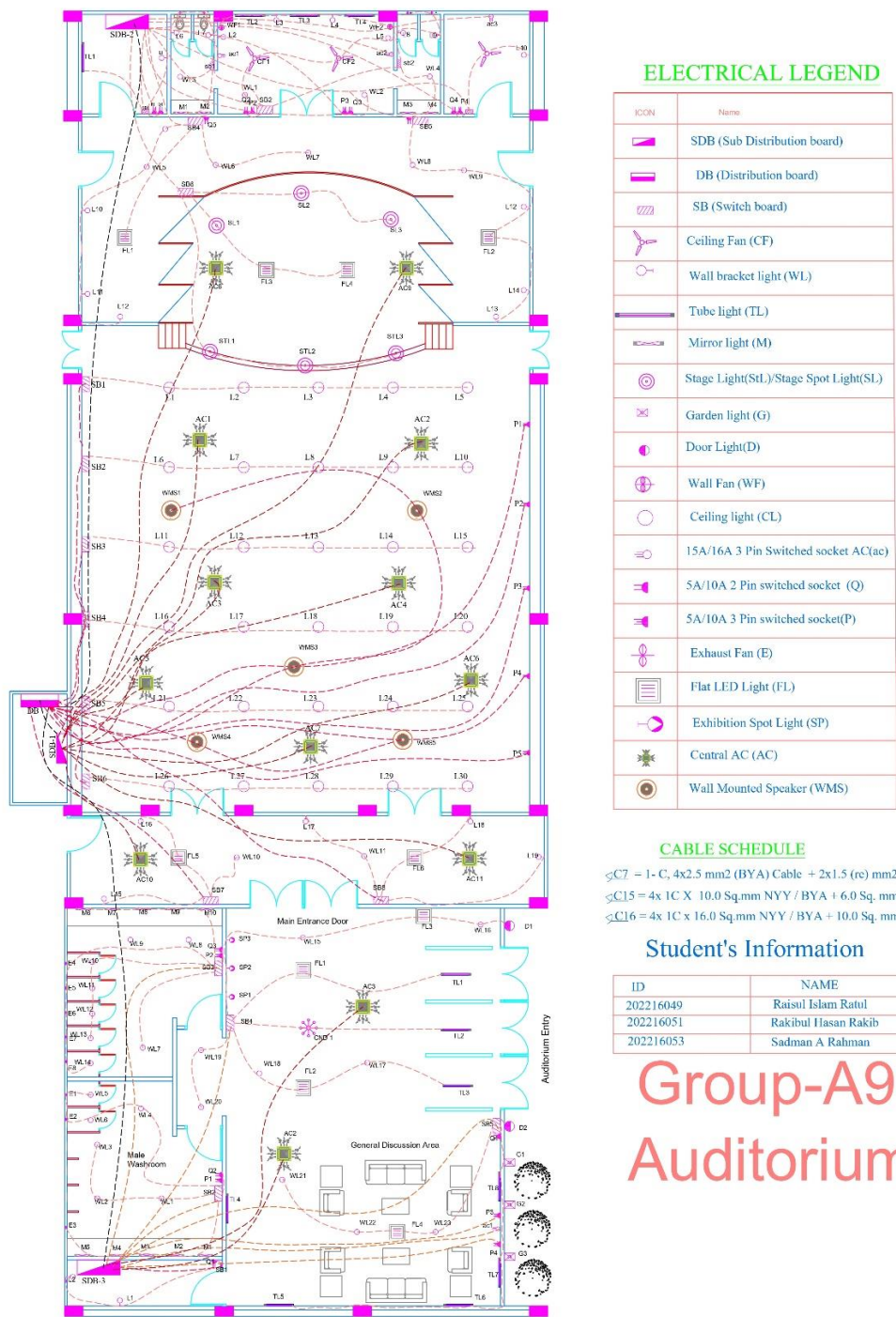


Figure 4 Conduit Layout

Figure 4 – SLD

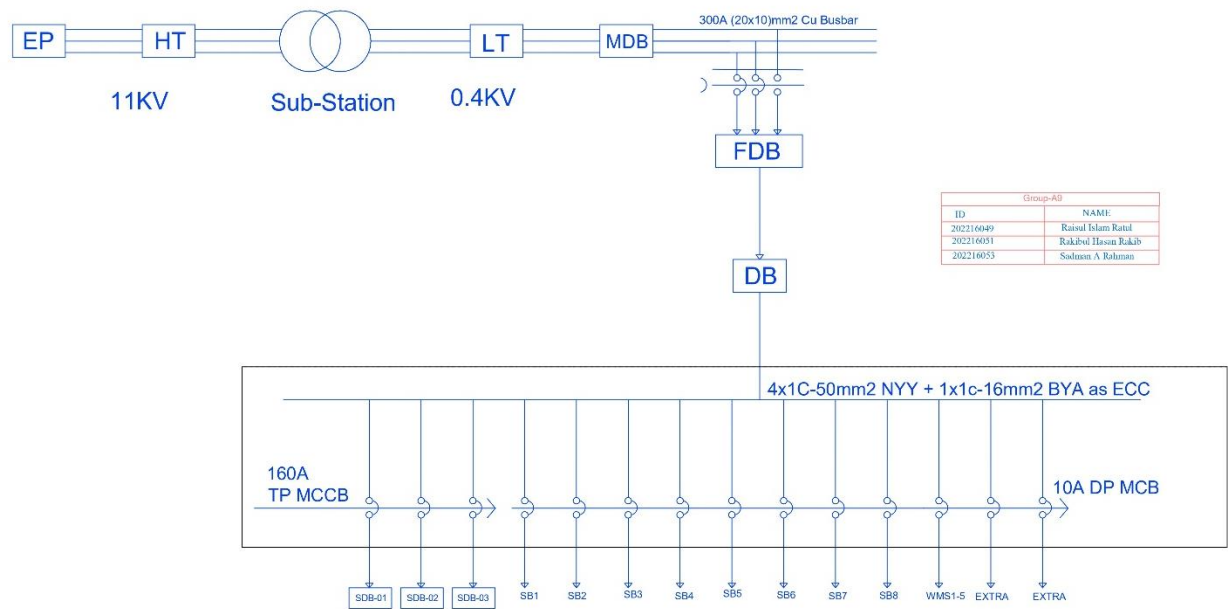
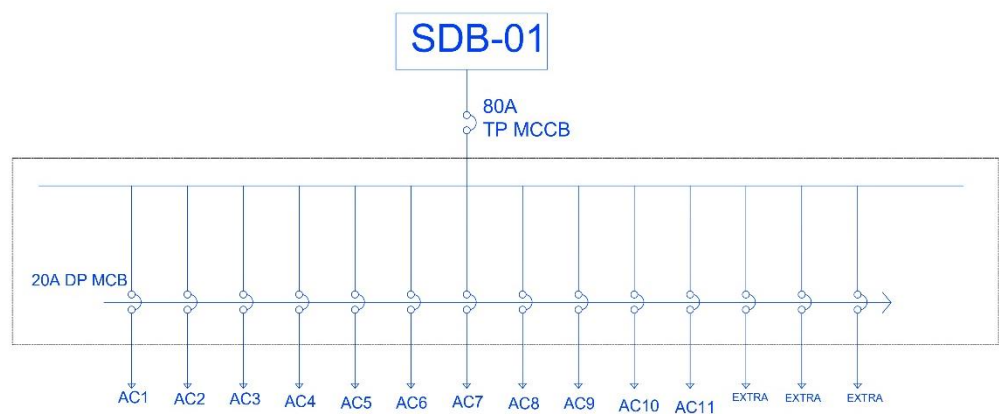


Figure 5 Substation to DB



Group-A9	
ID	NAME
202216049	Raisul Islam Ratal
202216051	Rakibul Hasan Rakib
202216053	Sadman A Rahman

Figure 6 SDB1

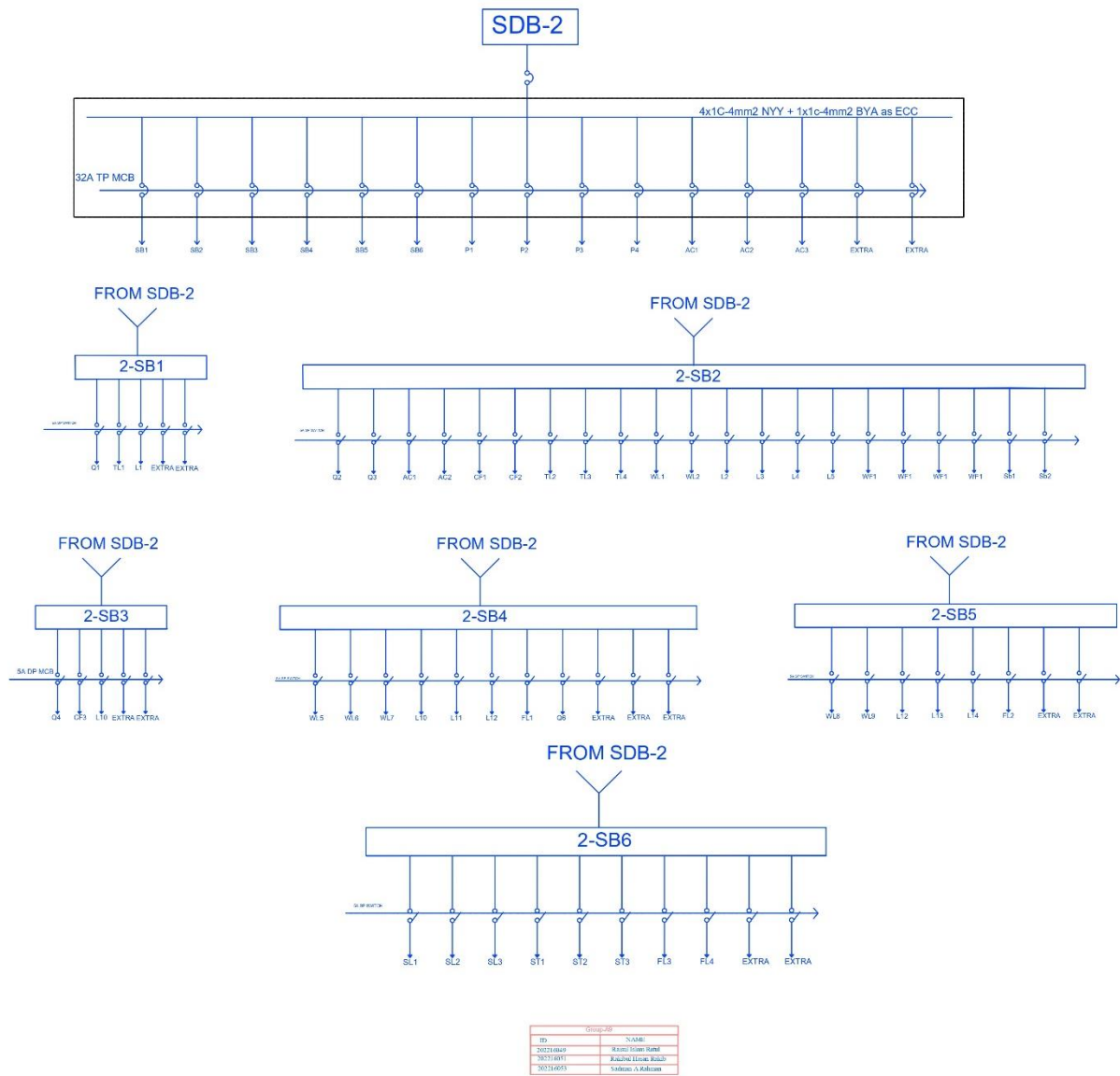


Figure 7 SDB2

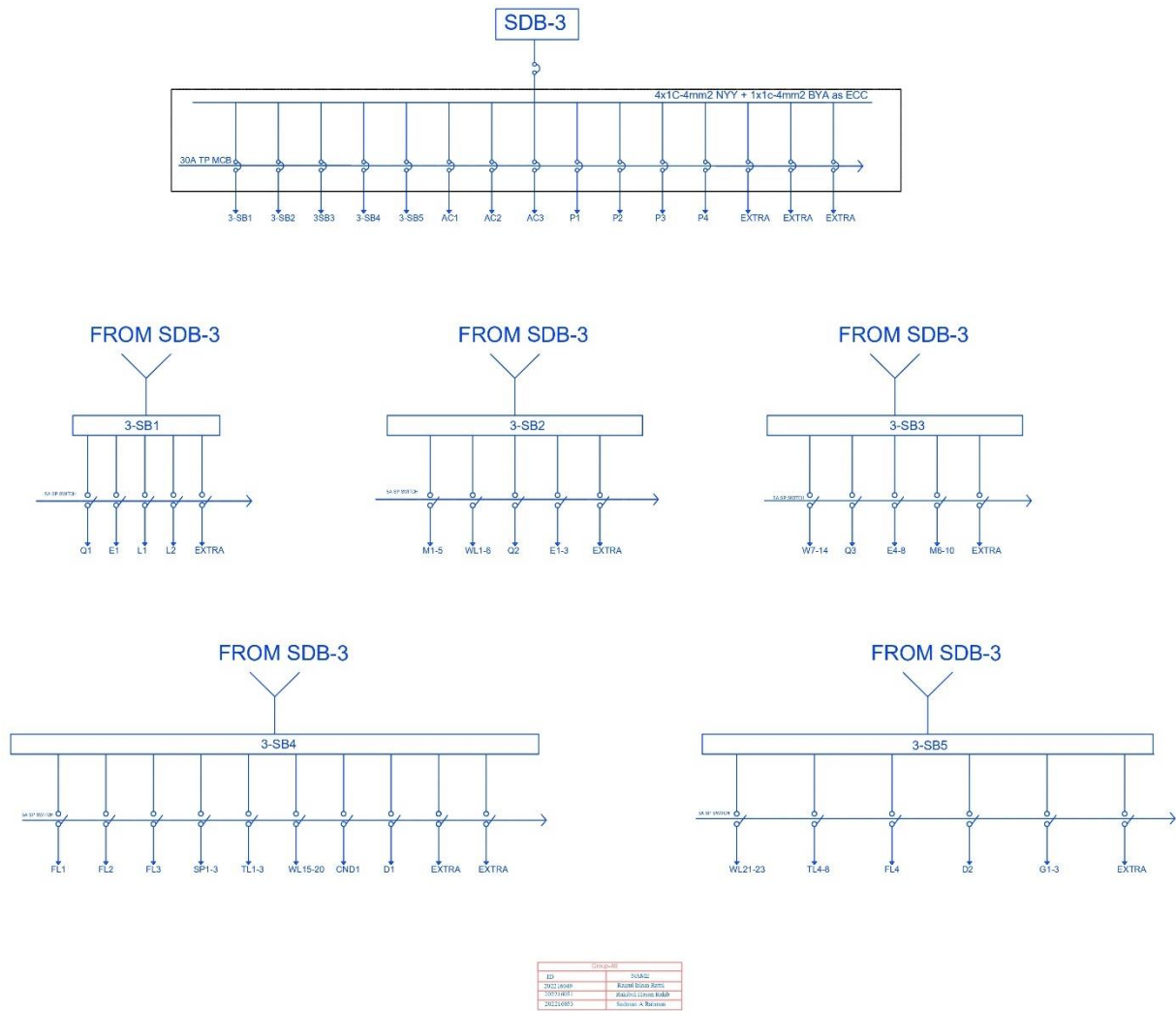


Figure 8 SDB3

Calculation:

a. Load calculation.

Load Calculation for Auditorium						
Distribution Board						
SL no	Descripting of Items	Considered Rating(Watt)	Quantit y	Total Wattage	D.F(%)	Maximum Demand (Watt)
i.	Lights					
	Circular LED Light	60	30	1800	60%	1080
	Flat LED Light	80	2	160	60%	96
	Wall bracket light (WL)	40	5	200	60%	120
	Ceiling light (CL)	40	2	80	60%	48
	13A -3pin (type-S)	1000	5	5000	60%	3000
	Wall Mounted Speaker	600	5	3000	100%	3000
	Total Maximum Demand (Including SDB1+SDB2+SDB3)					64314.1
	with 10% spare					70745.51
	Therefore, I					123.0269254
	with 25% of RF					153.7836567
	Circuit Breaker: 160A DP MCB					
	Cable: 4x1C-50mm² NYY + 1x1c-16mm² BYA as ECC					

Sub Distribution Board-01						
Serial No	Descripting of Items	Considered Rating(Watt)	Quantit y	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	AC Loads					
	Central AC	3000	11	33000	100%	33000
	Total Maximum Demand					33000
	with 10% spare					36300
	Therefore, I					63.12594809
	with 25% of RF					78.90743511
	Circuit Breaker: 80A DP MCB					
	Cable: 4x1C-25mm² NYY + 1x1c-16mm² BYA as ECC					

Sub Distribution Board-02						
Serial No	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	Lights					
	Flat LED Light	80	4	320	60%	192
	Mirror Light	15	4	60	60%	36
	Tube Light	40	5	200	60%	120
	Wall bracket light (WL)	40	16	640	60%	384
	Stage Spot Light	800	3	2400	60%	1440
	Incandescent Rotating Stage Lights	1500	3	4500	60%	2700
	Ceiling light (CL)	40	8	320	60%	192
ii.	Fan					
	Ceiling Fan	75	3	225	50%	112.5
	Wall Mounted Fan	45	2	90	50%	45
ii.	Power Sockets					
	5A-2pin	300	6	1800	50%	900
	13A-3pin	1000	5	5000	60%	3000
iii.	AC Loads					
	AC(1.5 TON)	1500	2	3000	100%	3000
iv.	Others					
	Washroom Exhaust Fan	36	4	144	60%	86.4
	Total Maximum Demand					12207.9
	with 10% spare					13428.69
	Therefore, I					23.35258369
	with 25% of RF					29.19072961
	Circuit Breaker: 32A DP MCB					
	Cable: 4x1C-4mm² NYY + 1x1c-4mm² BYA as ECC					

Sub Distribution Board-03						
Serial No	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	Lights					
	Chandelier	400	1	400	100%	400
	Flat LED Light	80	4	320	100%	320
	Exhibition Spot Light	15	3	45	60%	27
	Mirror Light	15	10	150	60%	90
	Tube Light	40	8	320	60%	192
	Wall bracket light (WL)	40	2	80	60%	48
	Ceiling light (CL)	40	23	920	60%	552
	Door Light	30	2	60	100%	60
	Security Light	50	1	50	100%	50
	Garden Light	40	3	120	60%	72
ii	Power Sockets					
	5A-2pin	300	2	600	50%	300
	13A-3pin	1000	4	4000	50%	2000
iii.	Others					
	Washroom Exhaust Fan	36	7	252	60%	151.2
iv.	AC Loads					
	Central AC	3000	2	6000	100%	6000
	AC(1.5 TON)	1500	1	1500	100%	1500
	Total Maximum Demand					11762.2
	with 10% spare					12938.42
	Therefore, I					22.50000081
	with 25% of RF					28.12500101
	Circuit Breaker : 32A DP MCB					
	Cable: 4x1C-4mm2 NYY + 1x1c-4mm2 BYA as ECC					

b. Bus bar size, Cable size and circuit breaker size determination

Bus Bar Size Chart

Serial No	Circuit Breaker	Bus bar Size (A)	Cable Size (mm ²) with ECC	
01	06	20	1.5	1.5
02	10	20	2.5	2.5
03	16	32	2.5	2.5
04	20	40	4.0	4.0
05	32	60	6.0	6.0
06	40	80	10	10
07	50	100	16	16
08	63	120	16	16
09	80	160	25	16
10	100	200	35	16
11	125	250	50	25
12	160	300	70	35
13	200	400	95	50
14	250	500	120	70
15	300	600	150	70
16	320	600	185	95
17	400	800	240	120

Figure 9 Bus bar chart

Bus Bar size is double than Circuit breaker

From Distribution Board to Sub- Distribution Board

From figure 9:

Circuit Breaker	Bus Bar Size(A)
160A DP MCB	300A



Applicaton: Suitable for indoors, outdoors, underground and in water for continuous permissible service voltage of 720/1200 Volts.



Construction: 1. Annealed Copper/Aluminium Conductor 2. PVC Insulation 3. Black PVC outer Sheath.

Colour: ●

PHYSICAL DATA							ELECTRICAL DATA						
Cross Sectional Area	No. & approx. dia. of wire Cu/Al	Shape of Conductor	Nominal thickness of insulation	Nominal thickness of sheath	Overall diameter (approx)	Approx. weight of cable		Max. DC resistance of conductor at 20 °C		Current rating at 30 °C in ground		Current rating at 35 °C in air	
						Cu	Al	Cu	Al	Cu	Al	Cu	Al
core x mm ²	nos./mm	-	mm	mm	mm	kg/km	kg/km	ohm/km	ohm/km	amps	amps	amps	amps
1x1.5	1/1.38	re	0.8	1.8	6.6	55	-	12.1	18.1	27	-	22	-
1x1.5	7/0.52	rm	0.8	1.8	6.8	58	-	12.1	18.1	27	-	22	-
1x2.5	1/1.78	re	0.8	1.8	7.2	70	-	7.41	12.1	36	-	30	-
1x2.5	7/0.68	rm	0.8	1.8	7.4	75	-	7.41	12.1	36	-	30	-
1 x 4	7/0.85	rm	1.0	1.8	8.2	106	80	4.61	7.41	47	37	39	31
1 x 6	7/1.04	rm	1.0	1.8	8.7	132	92	3.08	4.61	59	48	50	41
1x10	7/1.35	rm	1.0	1.8	9.7	182	118	1.83	3.08	78	60	69	53
1x16	7/1.70	rm	1.0	1.8	10.7	252	150	1.15	1.91	100	78	94	73
1x25	7/2.14	rm	1.2	1.8	12.4	363	202	0.727	1.2	130	101	125	97
1x35	min. 6	rm	1.2	1.8	13.7	470	248	0.524	0.868	155	120	160	124
1x50	min. 6	rm	1.4	1.8	15.6	645	332	0.387	0.641	185	144	195	151
1x70	min. 12	rm	1.4	1.8	17.3	858	420	0.268	0.443	225	175	245	190
1 x95	min. 15	rm	1.6	1.8	19.4	1129	530	0.193	0.32	270	210	300	232
1X120	min. 18/15	rm	1.6	1.8	21	1384	628	0.153	0.253	310	240	350	272
1x150	min. 18/15	rm	1.8	1.8	23.1	1709	780	0.124	0.206	350	270	405	314
1x185	min. 30	rm	2	2.0	25.6	2097	932	0.0991	0.164	390	302	460	357
1 x240	min. 34/30	rm	2.2	2.0	28.6	2708	1190	0.0754	0.125	450	349	555	430
1 x300	min. 34/30	rm	2.4	2.0	31.3	3405	1248	0.0601	0.1	515	386	640	448
1x400	min. 53	rm	2.6	2.2	35.3	4408	1952	0.047	0.0778	585	439	770	540
1 x500	min. 53	rm	2.8	2.2	38.0	5436	2340	0.0366	0.0605	680	510	900	630
1 x630	min. 53	rm	2.8	2.2	42.0	6780	2780	0.0283	0.0469	800	600	1030	721
1 x800	min. 53	rm	2.8	2.4	46.2	8510	3195	0.0221	0.0367	945	708	1160	812
1X1000	min. 53	rm	3.0	2.5	51.1	10530	3350	0.0176	0.0291	1095	821	1310	917

Total Maximum Demand(Including SDB1+SDB2+SDB3)	64314.1
with 10% spare	70745.51
Therefore, I	123.0269254
with 25% of RF	153.7836567
Cable: 4x1C-50mm2 NYY + 1x1c-16mm2 BYA as ECC	

For Sub – Distribution Board -1

From figure-9

Circuit Breaker	Bus Bar Size(A)
80A DP MCB	160A

NYY/NAYY
600/1000 V
SINGLE CORE
VDE 0271

Application: Suitable for indoors, outdoors, underground and in water for continuous permissible service voltage of 720/1200 Volts.



Construction: 1. Annealed Copper/Aluminium Conductor 2. PVC Insulation 3. Black PVC outer Sheath.

Colour: ●

PHYSICAL DATA						ELECTRICAL DATA							
Cross Sectional Area	No. & approx. dia. of wire	Shape of Conductor	Nominal thickness of insulation	Nominal thickness of sheath	Overall diameter (approx)	Approx. weight of cable		Max. DC resistance of conductor at 20 °C		Current rating at 30 °C in ground		Current rating at 35 °C in air	
core x mm ²	nos./mm	-	mm	mm	mm	Cu kg/km	Al kg/km	Cu ohm/km	Al ohm/km	Cu amps	Al amps	Cu amps	Al amps
1x1.5	1/1.38	re	0.8	1.8	6.6	55	-	12.1	18.1	27	-	22	-
1x1.5	7/0.52	rm	0.8	1.8	6.8	58	-	12.1	18.1	27	-	22	-
1x2.5	1/1.78	re	0.8	1.8	7.2	70	-	7.41	12.1	36	-	30	-
1x2.5	7/0.68	rm	0.8	1.8	7.4	75	-	7.41	12.1	36	-	30	-
1x4	7/0.85	rm	1.0	1.8	8.2	106	80	4.61	7.41	47	37	39	31
1x6	7/1.04	rm	1.0	1.8	8.7	132	92	3.08	4.61	59	48	50	41
1x10	7/1.35	rm	1.0	1.8	9.7	182	118	1.83	3.08	78	60	69	53
1x16	7/1.70	rm	1.0	1.8	10.7	252	150	1.15	1.91	100	78	94	73
1x25	7/2.14	rm	1.2	1.8	12.4	363	202	0.727	1.2	130	101	125	97
1x35	min. 6	rm	1.2	1.8	13.7	470	248	0.524	0.868	155	120	160	124
1x50	min. 6	rm	1.4	1.8	15.6	645	332	0.387	0.641	185	144	195	151
1x70	min. 12	rm	1.4	1.8	17.3	858	420	0.268	0.443	225	175	245	190
1x95	min. 15	rm	1.6	1.8	19.4	1129	530	0.193	0.32	270	210	300	232
1x120	min. 18/15	rm	1.6	1.8	21	1384	628	0.153	0.253	310	240	350	272
1x150	min. 18/15	rm	1.8	1.8	23.1	1709	780	0.124	0.206	350	270	405	314
1x185	min. 30	rm	2	2.0	25.6	2097	932	0.0991	0.164	390	302	460	357
1x240	min. 34/30	rm	2.2	2.0	28.6	2708	1190	0.0754	0.125	450	349	555	430
1x300	min. 34/30	rm	2.4	2.0	31.3	3405	1248	0.0601	0.1	515	386	640	448
1x400	min. 53	rm	2.6	2.2	35.3	4408	1952	0.047	0.0778	585	439	770	540
1x500	min. 53	rm	2.8	2.2	38.0	5436	2340	0.0366	0.0605	680	510	900	630
1x630	min. 53	rm	2.8	2.2	42.0	6780	2780	0.0283	0.0469	800	600	1030	721
1x800	min. 53	rm	2.8	2.4	46.2	8510	3195	0.0221	0.0367	945	708	1160	812
1x1000	min. 53	rm	3.0	2.5	51.1	10530	3350	0.0176	0.0291	1095	821	1310	917

Total Maximum Demand	33000
with 10% spare	36300
Therefore, I	63.12594809
with 25% of RF	78.90743511
Cable: 4x1C-25mm2 NYN + 1x1c-16mm2 BYA as ECC	

For Sub-Distribution Board-2

From figure 9:

Circuit Breaker:	Bus Bar Size(A)
32A DP MCB	60A

NYY/NAYY
600/1000 V
SINGLE CORE
VDE 0271

Application: Suitable for indoors, outdoors, underground and in water for continuous permissible service voltage of 720/1200 Volts.



Construction: 1. Annealed Copper/Aluminium Conductor 2. PVC Insulation 3. Black PVC outer Sheath.

Colour: ●

PHYSICAL DATA							ELECTRICAL DATA						
Cross Sectional Area	No. & approx. dia. of wire Cu/Al	Shape of Conductor	Nominal thickness of insulation	Nominal thickness of sheath	Overall diameter (approx)	Approx. weight of cable		Max. DC resistance of conductor at 20 °C		Current rating at 30 °C in ground		Current rating at 35° C in air	
						Cu	Al	Cu	Al	Cu	Al	Cu	Al
core x mm ²	nos./mm	-	mm	mm	mm	kg/km	kg/km	ohm/km	ohm/km	amps	amps	amps	amps
1x1.5	1/1.38	re	0.8	1.8	6.6	55	-	12.1	18.1	27	-	22	-
1x1.5	7/0.52	rm	0.8	1.8	6.8	58	-	12.1	18.1	27	-	22	-
1x2.5	1/1.78	re	0.8	1.8	7.2	70	-	7.41	12.1	36	-	30	-
1x2.5	7/0.68	rm	0.8	1.8	7.4	75	-	7.41	12.1	36	-	30	-
1x4	7/0.85	rm	1.0	1.8	8.2	106	80	4.61	7.41	47	37	39	31
1x6	7/1.04	rm	1.0	1.8	8.7	132	92	3.08	4.61	59	48	50	41
1x10	7/1.35	rm	1.0	1.8	9.7	182	118	1.83	3.08	78	60	69	53
1x16	7/1.70	rm	1.0	1.8	10.7	252	150	1.15	1.91	100	78	94	73
1x25	7/2.14	rm	1.2	1.8	12.4	363	202	0.727	1.2	130	101	125	97
1x35	min. 6	rm	1.2	1.8	13.7	470	248	0.524	0.868	155	120	160	124
1x50	min. 6	rm	1.4	1.8	15.6	645	332	0.387	0.641	185	144	195	151
1x70	min. 12	rm	1.4	1.8	17.3	858	420	0.268	0.443	225	175	245	190
1x95	min. 15	rm	1.6	1.8	19.4	1129	530	0.193	0.32	270	210	300	232
1X120	min. 18/15	rm	1.6	1.8	21	1384	628	0.153	0.253	310	240	350	272
1x150	min. 18/15	rm	1.8	1.8	23.1	1709	780	0.124	0.206	350	270	405	314
1x185	min. 30	rm	2	2.0	25.6	2097	932	0.0991	0.164	390	302	460	357
1x240	min. 34/30	rm	2.2	2.0	28.6	2708	1190	0.0754	0.125	450	349	555	430
1x300	min. 34/30	rm	2.4	2.0	31.3	3405	1248	0.0601	0.1	515	386	640	448
1x400	min. 53	rm	2.6	2.2	35.3	4408	1952	0.047	0.0778	585	439	770	540
1x500	min. 53	rm	2.8	2.2	38.0	5436	2340	0.0366	0.0605	680	510	900	630
1x630	min. 53	rm	2.8	2.2	42.0	6780	2780	0.0283	0.0469	800	600	1030	721
1x800	min. 53	rm	2.8	2.4	46.2	8510	3195	0.0221	0.0367	945	708	1160	812
1X1000	min. 53	rm	3.0	2.5	51.1	10530	3350	0.0176	0.0291	1095	821	1310	917

Total Maximum Demand	12207.9
with 10% spare	13428.69
Therefore, I	23.35258369
with 25% of RF	29.19072961
Cable: 4x1C-4mm ² NYN + 1x1c-4mm ² BYA as ECC	

For Sub-Distribution Board-03

From Figure 9:

Circuit Breaker	Bus Bar Size(A)
32A DP MCB	300A



Application: Suitable for indoors, outdoors, underground and in water for continuous permissible service voltage of 720/1200 Volts.



Construction: 1. Annealed Copper/Aluminium Conductor 2. PVC Insulation 3. Black PVC outer Sheath.

Colour: ●

PHYSICAL DATA							ELECTRICAL DATA						
Cross Sectional Area	No. & approx. dia. of wire Cu/Al	Shape of Conductor	Nominal thickness of insulation	Nominal thickness of sheath	Overall diameter (approx)	Approx. weight of cable	Max. DC resistance of conductor at 20 °C	Current rating at 30 °C in ground	Current rating at 35 °C in air				
core x mm ²	nos./mm	-	mm	mm	mm	Cu kg/km	Al kg/km	Cu ohm/km	Al ohm/km	Cu amps	Al amps	Cu amps	Al amps
1x1.5	1/1.38	re	0.8	1.8	6.6	55	-	12.1	18.1	27	-	22	-
1x1.5	7/0.52	rm	0.8	1.8	6.8	58	-	12.1	18.1	27	-	22	-
1x2.5	1/1.78	re	0.8	1.8	7.2	70	-	7.41	12.1	36	-	30	-
1x2.5	7/0.68	rm	0.8	1.8	7.4	75	-	7.41	12.1	36	-	30	-
1x4	7/0.85	rm	1.0	1.8	8.2	106	80	4.61	7.41	47	37	39	31
1x6	7/1.04	rm	1.0	1.8	8.7	132	92	3.08	4.61	59	48	50	41
1x10	7/1.35	rm	1.0	1.8	9.7	182	118	1.83	3.08	78	60	69	53
1x16	7/1.70	rm	1.0	1.8	10.7	252	150	1.15	1.91	100	78	94	73
1x25	7/2.14	rm	1.2	1.8	12.4	363	202	0.727	1.2	130	101	125	97
1x35	min. 6	rm	1.2	1.8	13.7	470	248	0.524	0.868	155	120	160	124
1x50	min. 6	rm	1.4	1.8	15.6	645	332	0.387	0.641	185	144	195	151
1x70	min. 12	rm	1.4	1.8	17.3	858	420	0.268	0.443	225	175	245	190
1x95	min. 15	rm	1.6	1.8	19.4	1129	530	0.193	0.32	270	210	300	232
1X120	min. 18/15	rm	1.6	1.8	21	1384	628	0.153	0.253	310	240	350	272
1x150	min. 18/15	rm	1.8	1.8	23.1	1709	780	0.124	0.206	350	270	405	314
1x185	min. 30	rm	2	2.0	25.6	2097	932	0.0991	0.164	390	302	460	357
1x240	min. 34/30	rm	2.2	2.0	28.6	2708	1190	0.0754	0.125	450	349	555	430
1x300	min. 34/30	rm	2.4	2.0	31.3	3405	1248	0.0601	0.1	515	386	640	448
1x400	min. 53	rm	2.6	2.2	35.3	4408	1952	0.047	0.0778	585	439	770	540
1x500	min. 53	rm	2.8	2.2	38.0	5436	2340	0.0366	0.0605	680	510	900	630
1x630	min. 53	rm	2.8	2.2	42.0	6780	2780	0.0283	0.0469	800	600	1030	721
1x800	min. 53	rm	2.8	2.4	46.2	8510	3195	0.0221	0.0367	945	708	1160	812
1X1000	min. 53	rm	3.0	2.5	51.1	10530	3350	0.0176	0.0291	1095	821	1310	917

Total Maximum Demand	11762.2
with 10% spare	12938.42
Therefore, I	22.50000081
With 25% of RF	28.12500101
Cable: 4x1C-4mm2 NYY + 1x1c-4mm2 BYA as ECC	

c. Illumination calculation

We have to calculate the number of light needed for main hall area.

We have,

Illuminance, $E = 200$

Maintenance factor, $mf = 0.7$

Utilization Factor, $uf = 0.6$

Average flux from each light, $F = 2600$

Length, $L = 39\text{ ft } 2\text{ inch} = 11.8872\text{m}$

Width = $42\text{ ft. } 7\text{ inch} = 13.1064\text{m}$

Area = $L \times W = 155.59\text{m}^2$

$$N = \frac{EA}{F \times mf \times uf}$$
$$N = \frac{200 \times 155.59}{2600 \times 0.7 \times 0.6}$$
$$= 28.49 \approx 30$$

Mounting Height, $H_m = 10\text{m}$

Space to Height Ratio, $SHR = \frac{1}{10} \times \sqrt{\frac{155.59}{28}} = 0.235$

Now,

$$\frac{s}{H_m} = 0.235$$
$$s = 2.4$$

Number of lights in each Row = $\frac{11.887}{2.4} = 4.9 \approx 5$

Number of lights in each Column= $\frac{28}{5} = 5.6 \approx 6$

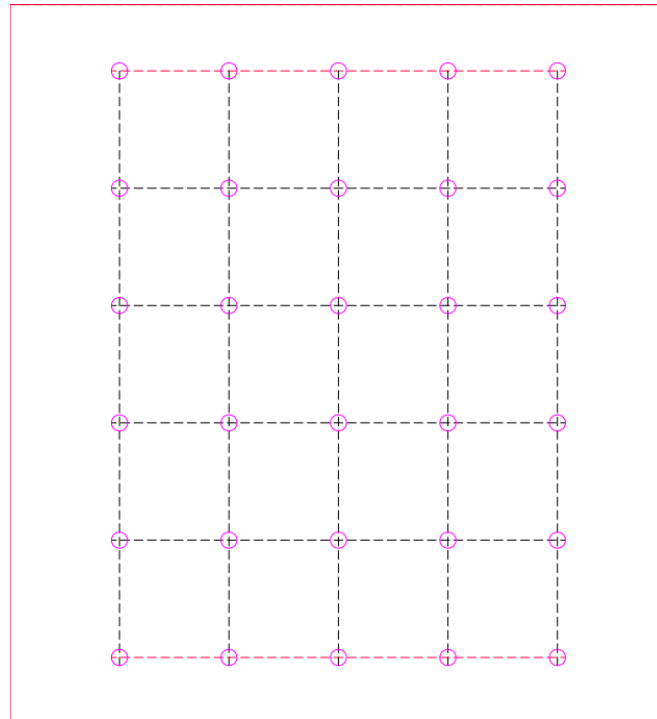


Figure 10 Illumination Diagram

Estimation of Total Cost

BOQ Calculation for Auditorium					
Item No	Description	Unit	Quantity	Unit Rate(BDT)	Total Amount
1	Cable works				
	4x1C-50mm2 NYY(BRB Cables)	Meter	176	3,383.83	595,554.08
	4x1C-25mm2 NYY(Bizli Cables)	Meter	1250	2,116.40	2,645,500.00
	4x1C-4mm2 NYY	Meter	321	415.40	133,343.40
	4x1C-4mm2 NYY	Meter	66	415.40	27,416.40
2	Cable works (ECC)				
	1x1c-16mm2 BYA	Meter	180	196.40	35,352.00
	1x1c-16mm2 BYA	Meter	1270	196.40	249,428.00
	1x1c-4mm2 BYA	Meter	330	25.58	8,441.40
	1x1c-4mm2 BYA	Meter	75	25.58	1,918.50
3	Circuit Breakers				
	160A TP MCCB	Each	3	7,500.00	22,500.00
	80A DP MCB	Each	15	1,200.00	18,000.00
	32A DP MCB	Each	15	310.00	4,650.00
	32A DP MCB	Each	14	310.00	4,340.00
	10A DP MCB	Each	11	260.00	2,860.00
	20A TP MCB	Each	14	427.00	5,978.00
4	Air condition				
	Central AC	Each	13	264,000.00	3,432,000.00
	1.5 TON AC	Each	4	57,900.00	231,600.00
5	Sound System				-
	Wall Mounted Speaker	Each	5	14,000.00	70,000.00
6	Busbar				

	300A (20x10)mm ² Cu Busbar	meter	6	2,700.00	16,200.00
7	Lights				
	Tube light (TL)	Each	15	1,220.00	18,300.00
	Mirror light (M)	Each	14	620.00	8,680.00
	Stage Spot Light(SL)		3	11,577.00	34,731.00
	Garden light (G)		3	389.00	1,167.00
	Door Light(D)		2	189.00	378.00
	Exhibition Spot Light (SP)		3	621.00	1,863.00
	Flat LED Light (FL)		10	2,400.00	24,000.00
	Ceiling light (CL)		35	1,450.00	50,750.00
	Stage Light(StL)		3	17,500.00	52,500.00
8	Fans				
	1440 mm ² ceiling fan	Each	4	2,000.00	8,000.00
9	Sockets				
	3 Pin 13 A socket	Each	10	375.00	3,750.00
	2 pin 5A socket	Each	13	35.00	455.00
10	Switch's				
	1 way 4 gang 10A switch	Each	156	400.00	62,400.00
11	Conduit				
	Minimum inner Dia. 20 mm & minimum wall thickness 1.5 mm.	meter	1426	40.00	57,040.00
	Minimum inner Dia. 25 mm & minimum wall thickness 1.5 mm.	meter	421	55.00	23,155.00
	Total Cost				7,852,250.78

By performing these calculations diligently, one can accurately determine the electrical load, specify the required electrical components, design appropriate lighting systems, and provide a reasonable estimation of the total cost for an Auditorium construction project. Professional consultation and adherence to relevant electrical standards are crucial throughout these processes for accurate and safe installations.

Conclusion:

The Auditorium's floor plan exemplifies a harmonious convergence of efficiency, suitability, and cost-effectiveness, crafting a versatile space designed to cater to diverse events while emphasizing functionality and economic viability.

Careful planning and layout orchestration of the main hall, stage area, technical booths, and ancillary rooms ensure a seamless experience for presenters and audiences while meeting safety and accessibility standards. This thoughtful arrangement optimizes event operations, ensuring optimal visibility, acoustics, and technical support.

Efficiency is deeply ingrained in the design's core. Precise load calculations and selection of electrical components guarantee an efficient system meeting demand loads and allowing for future expansions. Additionally, the lighting design, meticulously calculated to fulfill requirements while conserving energy, underscores the Auditorium's sustainability.

Cost-effectiveness is evident through prudent material selection, labor estimation, and project management. The design strikes a balance between functionality and economic feasibility, utilizing resources efficiently.

In summary, the Auditorium's floor plan harmonizes appropriateness, efficiency, and cost-effectiveness, serving as a model for versatile and economically feasible architectural designs.

Reference:

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